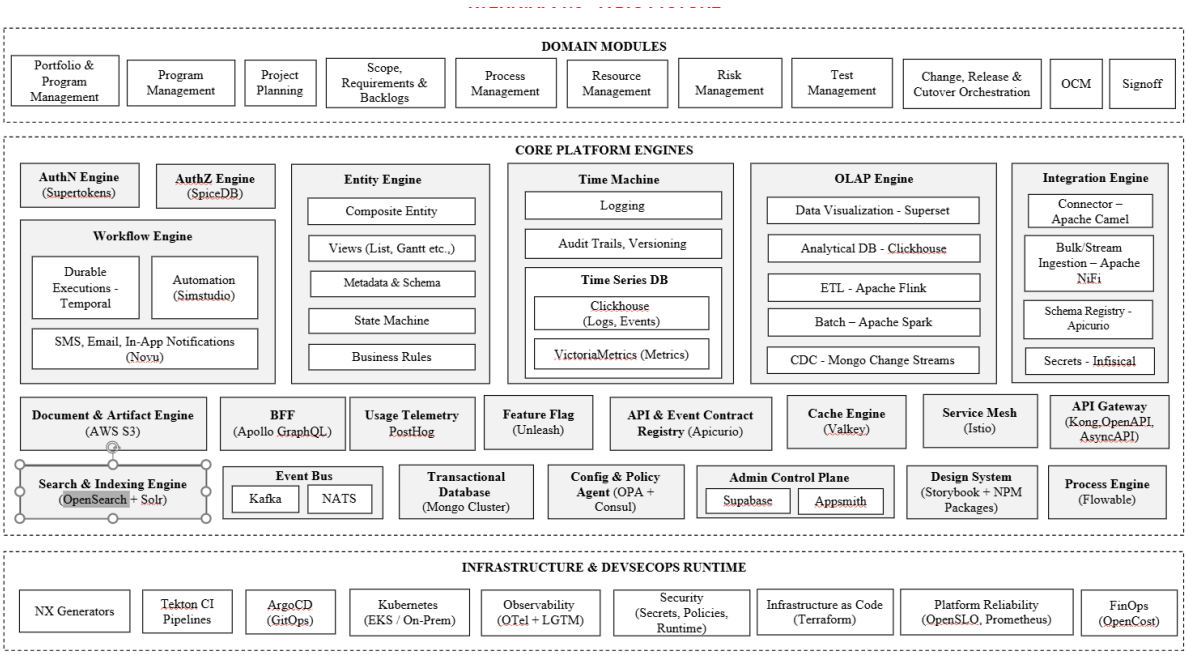
**Current KTern Application**

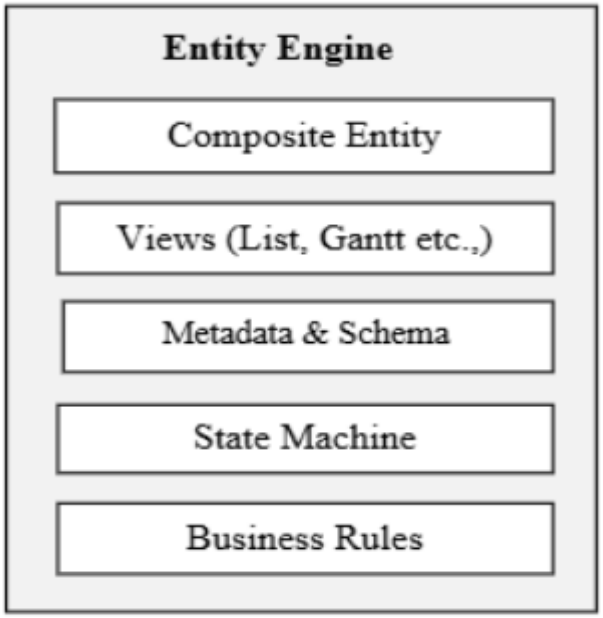
1. Docs - KTern.AI Documentation
2. Digital Projects - KTern.AI Documentation
3. Custom Boards - KTern.AI Documentation
4. How to create Custom Boards and tasks? - KTern.AI Documentation

Ktern v1.5 - Architecture (In Design and Development stage)




## Entity Engine – Main Focus

This replaces the Custom Boards(v1)



**High Level Idea/Use case**

Entity engine is a way where the customers can customize their requirements based on the use.

**Current Scenario – Problem Statement**

For example, any Project Management need to have these core domains which is often called PMO-Project Management Office

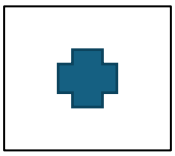
- Portfolio & Program Management
- Program Management
- Project Planning
- Scope, Requirement and Backlogs
- Process Management
- Resource Management
- Risk Management
- Test Management
- Change, Releases & Cutover orchestrator
- OCM framework
- Sign off

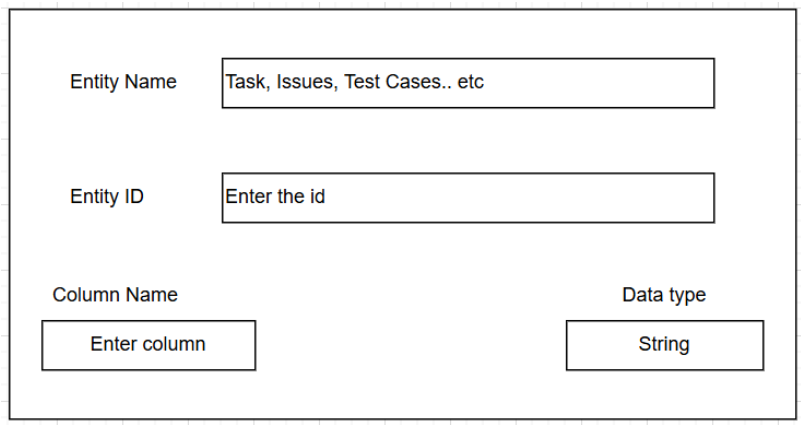
Our previous application contained hard-coded way ie by mistakely way to add task, issues, or testcases. And each column heading such as for Tasks for example has Title, Phase, Work package, Status [New, Active, Completed, approved], Priority, Planned From, planned to, completed on, Assigned to, Team, Task type, etc. And these columns are not constant, each customer requires different column title based on its needs. So the developer needs to hardcode each and every time a customer requests for different column title based on there requirement.

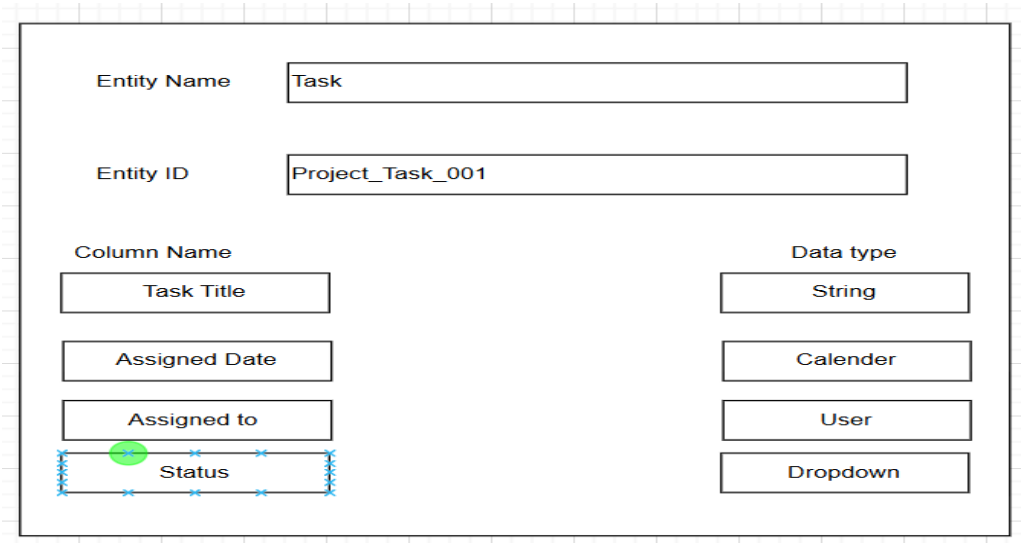
**Solution**

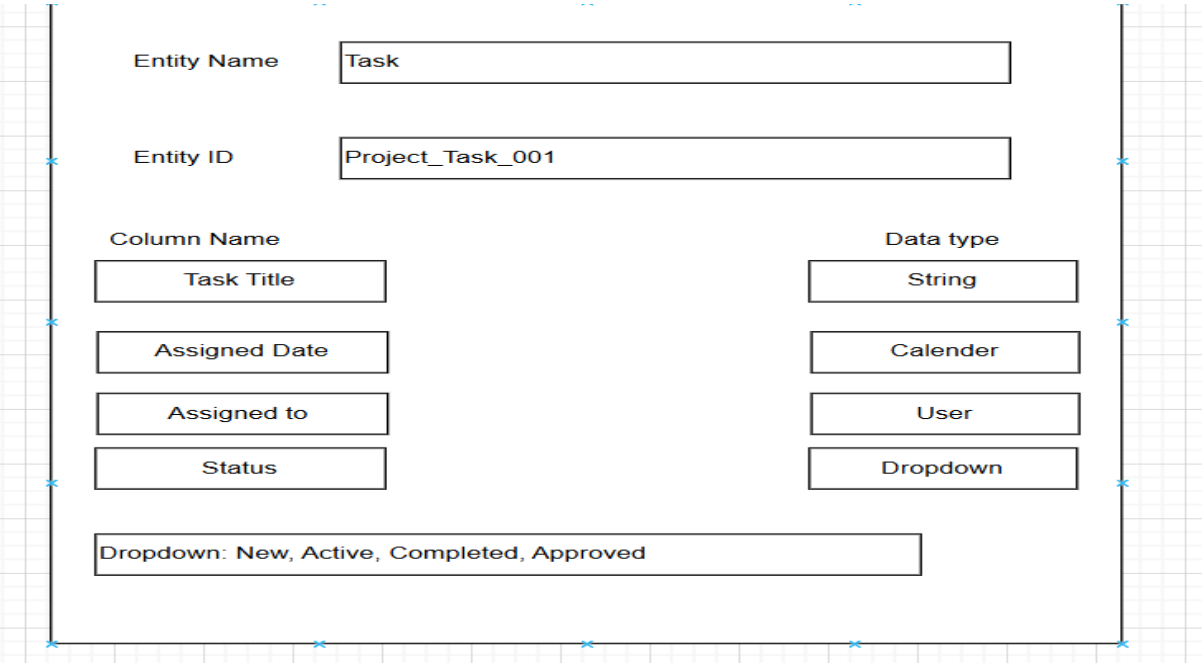
To overcome such hardcoding for different customer we have the idea of creating a **Entity Engine**

Let's see the feature/use of entity engine.

- A customizable way where the ui is like  Create entity.
- Customer can give the entity name as for example Task (as of now the customer need to add/ organize multiple line entry of tasks to do in a project)




This is a sample UI. Based on the customer requirement the customer can add the column title, and its required data type.

Example of Data type -> shows in drop down list. For example String, calendar, user.

- And then customer will click create then Entity Named – Task based on the customer required column tile is created.
- **Composite Entity** -> This is a feature where for example under a project, we can create/add.
  Task Entity | Requirement Entity| Issues Entity| Risk Entity| TestCases Entity|
  All these are related to one single project. So they can group all of them together under one entity to form a Composite Entity.
- **Views** -> List, Kanban, Calender
  The created entity can be shown in 3 views as of now that is list view, kanban view and calendar view to keep track of start and end date of the line entry, the duration etc.
- **Status** -> This shows in phase the task is / task state. For example when a task or any entity is created its created in NEW state.
- **State Machine** -> To make sure the state should change from New-> Active-> Completed-> Approved. This rule is followed using the State machine. This make sure that it should follow the above flow and no can come from new ro approved directly.
- There is something called Rule Engine where it has functions for mathematical calculations.

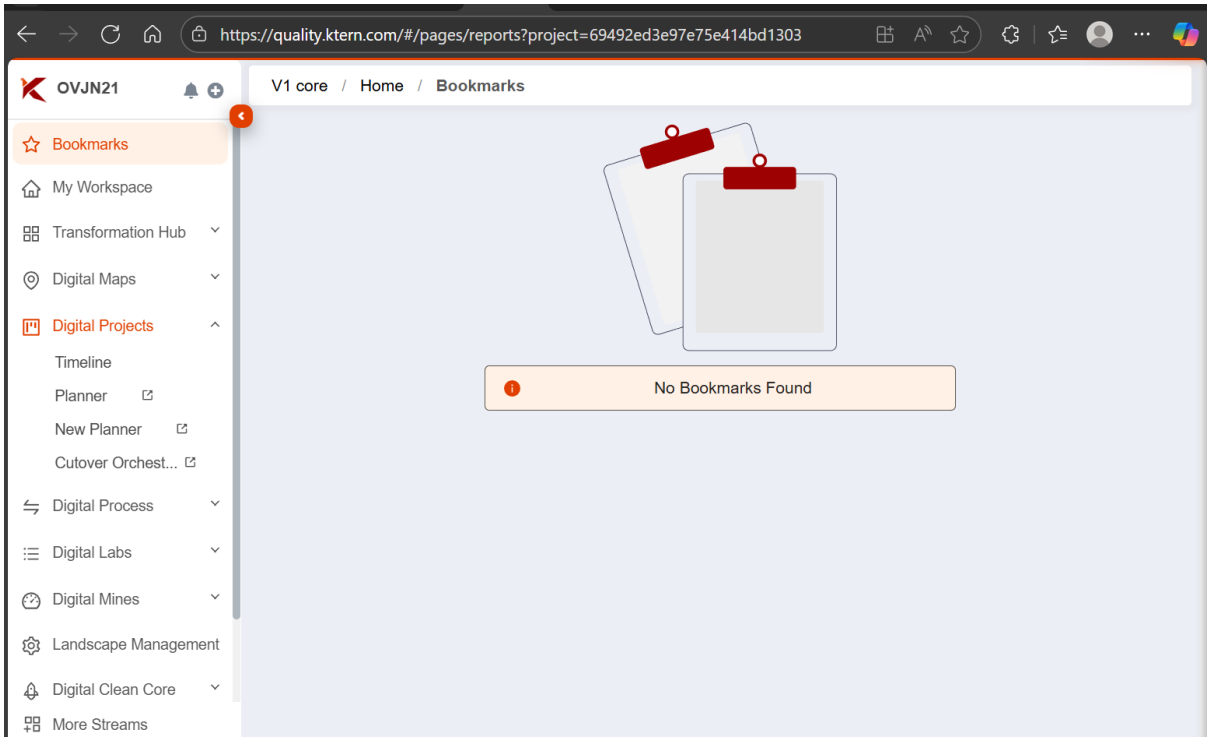
So this Entity Engine will be replaced with the existing application for custom boards.

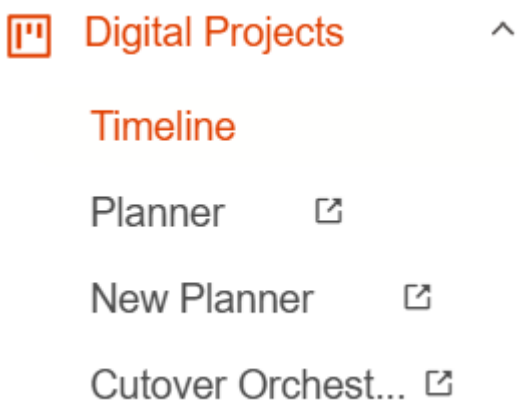
This is main idea behind the entity engine.

**Summary**

- Different customer request for different column titles based on the requirement.
- So far developers had to hardcode each and every change the different customers' requirement in the codebase.
- This became a cumbersome process for the developers, as it was difficult to keep track of each change.
- This in turn delayed the release which affected customer satisfaction.
- To make developers work easy and good customer satisfaction, introduced an core engine called Entity Engine – which provides a customizable way for customers to create entity and add there own required column titles based on their requiremet for any project management framework be it Tasks, isues, test cases, risks, etc.
- We are going to integrate this Entity Engine as a separate microservices in the current Ktern Application.

I have attached currrent Application documentation as url. Let me give the UI of the current application  launchpad for ur better understanding.




Under the Digital Projects we will have a Entity Engine if we click the arrow, it will open/ run in separate tab – making it a microservices.